Network

What are the two main components of the network layer?
a) Data plane and Control plane
b) Transport layer and Link layer
c) Forwarding and Buffering
d) IP addressing and NAT

The network layer service model for the Internet is:
a) Guaranteed delivery with fixed delay
b) Best effort
c) Constant bit rate
d) Guaranteed bandwidth

Which of the following describes the control plane?
a) It operates locally on each router.
b) It determines the routing path using network-wide logic.
c) It schedules packet transmission at the link layer.
d) It monitors physical layer signals.

Which layer encapsulates transport segments into datagrams?
a) Physical layer
b) Transport layer
c) Network layer
d) Application layer

Forwarding is responsible for:
a) Encapsulating data into frames
b) Determining the route from source to destination
c) Moving packets from input to output within a router
d) Assigning IP addresses dynamically

Which analogy best describes routing in the network layer?
a) Sending a letter via postal service
b) Sorting letters at a post office
c) Planning a trip from source to destination
d) Driving through a single traffic light

In router architecture, what operates in a nanosecond timeframe?
a) Control plane
b) Routing processor
c) Forwarding data plane
d) Buffer management

What is the purpose of a forwarding table in a router?
a) To prioritize packets based on size
b) To store active TCP connections
c) To determine the output port for incoming packets
d) To log all transmitted packets

Which of the following describes switching via a bus?
a) A datagram is sent directly to the destination router.
b) The datagram travels through a shared bus to reach the output port.
c) Packets are processed in parallel using multiple paths.
d) All incoming data is temporarily stored in a central memory.

What does "TTL" in an IP header stand for?
a) Total Transmission Loss
b) Transmission Time Log
c) Time to Live
d) Temporary Transmission Layer
How many bits are in an IPv4 address?
a) 16
b) 32
c) 48
d) 128
What is one major difference between IPv4 and IPv6?
a) IPv6 uses 64-bit addresses.
b) IPv6 provides built-in encryption.
c) IPv6 uses hexadecimal representation.
d) IPv6 has no support for NAT.
What causes "Head-of-Line (HOL) blocking" in input port queuing?
a) A datagram at the front of the queue blocks others.
b) Packets are misrouted due to errors in the forwarding table.
c) The output buffer overflows, causing packet loss.
d) The switch fabric is idle while input buffers fill.
What happens if a router's output buffer is full?
a) It drops all incoming packets.
b) It reduces the packet's time-to-live (TTL).
c) It selectively drops packets based on a drop policy.
d) It forwards packets at a higher rate.
What is the function of the router's switching fabric?
a) To schedule outgoing packets
b) To transfer packets from input ports to output ports
c) To manage routing tables
d) To adjust packet priorities dynamically

In packet scheduling, what does "Weighted Fair Queuing" (WFQ) provide?
a) Equal bandwidth to all flows
b) Bandwidth proportional to flow weights
c) Priority to the largest flows
d) The shortest delay for all packets
What does CIDR stand for?
a) Classless Internet Domain Routing
b) Classful Inter-Domain Routing
c) Classless Inter-Domain Routing
d) Connectionless Internet Domain Routing

In CIDR notation, what does "/24" mean?
a) The subnet mask has 24 bits.
b) The IP address is 24 bits long.
c) The host part of the address is 24 bits.
d) The address space includes 24 hosts.
How does DHCP assign IP addresses?
a) By using a static assignment method
b) By allowing the client to choose its address
c) By dynamically leasing addresses to devices
d) By mapping MAC addresses to IP addresses
Which of the following is NOT a benefit of DHCP?
a) Simplifies IP address management
b) Supports mobile users joining the network
c) Guarantees IP address persistence
d) Allows address reuse
What is the primary purpose of NAT?
a) To encrypt packets for secure transmission
b) To convert private IP addresses to a single public IP address
c) To assign dynamic IP addresses to devices
d) To manage routing tables in a network
Which of the following is a disadvantage of NAT?
a) It simplifies address configuration.
b) It introduces security vulnerabilities.
c) It violates the end-to-end principle of the Internet.
d) It requires IPv6 for implementation.
Which protocol resolves NAT traversal issues?
a) DHCP
b) ICMP
c) STUN
d) ARP
What is the size of an IPv6 address?
a) 32 bits
b) 48 bits
c) 64 bits
d) 128 bits
What feature was removed in IPv6 to speed up processing at routers?
a) Header checksum
b) Time-to-live field
c) Flow label
d) Source address
What is a common technique used to transition from IPv4 to IPv6?
a) Fragmentation
b) Tunneling
c) NAT traversal
d) Encapsulation

In generalized forwarding, what does the "match + action" abstraction involve?
a) Matching packet headers and determining routing paths
b) Matching packets to predefined policies and applying actions
c) Comparing packet payloads for integrity
d) Setting time limits for packet processing
Which field is commonly used for matching in SDN flow tables?
a) Packet payload
b) Destination IP address
c) Protocol version
d) Encapsulation type
In SDN, what is the role of a remote controller?
a) To queue packets based on priorities
b) To compute and install forwarding tables in routers
c) To manage physical-layer connections
d) To adjust TCP window sizes
Which packet scheduling method ensures packets are sent in the order they arrive?
a) Priority Scheduling
b) Weighted Fair Queuing
c) First Come, First Serve
d) Round Robin
What distinguishes priority scheduling from other methods?
a) It provides equal bandwidth to all flows.
b) It transmits packets based on arrival times.
c) It selects packets from the highest-priority queue first.
d) It minimizes queueing delay for all packets.
Which of the following is a feature of round-robin scheduling?
a) It processes packets in order of size.
b) It assigns time slots cyclically to each queue.
c) It guarantees no packet loss.
d) It prioritizes small packets.
What type of service does the Internet's "best-effort" model provide?
a) Guaranteed bandwidth
b) Reliable delivery with low latency
c) No guarantees for delivery, order, or timing
d) Delivery with QoS guarantees
Which service model is used for real-time applications with minimal delay?
a) Available Bit Rate
b) Best Effort
c) Constant Bit Rate
d) Datagram Delivery

What is the primary function of a middlebox?
a) To route packets based on IP addresses
b) To perform specialized processing beyond traditional IP routing
c) To manage link-layer connectivity
d) To assign IP addresses dynamically

Which of the following is NOT an example of a middlebox?
a) NAT device
b) Load balancer
c) Firewall
d) IP router
What is one advantage of using network functions virtualization (NFV) in middleboxes?
a) It provides hardware acceleration for packet forwarding.
b) It allows middlebox functions to run on general-purpose hardware.
c) It reduces the need for flow tables.
d) It simplifies physical-layer management.
Answer: b) It allows middlebox functions to run on general-purpose hardware.

What is the purpose of IP fragmentation?
a) To split datagrams into smaller units to fit the MTU of a link
b) To compress packets for faster transmission
c) To route packets through multiple paths
d) To convert IPv4 packets into IPv6 packets
What happens when a datagram's TTL reaches zero?
a) The datagram is sent back to the source.
b) The datagram is discarded by the router.
c) The router requests a new TTL value.
d) The router forwards the datagram with a warning.
What component does SDN use to enforce "match + action" policies?
a) Routing algorithms
b) Flow tables
c) Switch fabric
d) Transport protocols
What is the primary goal of packet scheduling in a router?
a) To maintain equal packet size across flows
b) To decide the transmission order of packets in a queue
c) To drop all low-priority packets during congestion
d) To prioritize packets based on source IP
Which packet scheduling method guarantees a minimum bandwidth for each traffic class?
a) Priority Scheduling
b) First Come, First Serve
c) Weighted Fair Queuing
d) Round Robin
What is a drawback of priority scheduling?
a) It treats all packets equally, regardless of priority.
b) It can lead to starvation of lower-priority packets.
c) It requires packet reassembly at the destination.
d) It is incompatible with IPv6.

Which IP address ranges are designated as private and used in NAT?
a) 192.0.2.0/24
b) 10.0.0.0/8
c) 240.0.0.0/4
d) 224.0.0.0/4
Which NAT variant maps one private IP to one public IP?
a) Port Address Translation (PAT)
b) Static NAT
c) Dynamic NAT
d) Overlapping NAT
What is a key advantage of NAT for networks?
a) Reduces router configuration complexity
b) Increases the number of public IP addresses available
c) Provides security by masking internal network addresses
d) Guarantees faster routing of packets
How does IPv6 improve routing efficiency compared to IPv4?
a) By using fixed-length 32-bit addresses
b) By removing the header checksum field
c) By introducing smaller headers
d) By restricting multicast communication
Which transition mechanism allows IPv6 packets to traverse IPv4 networks?
a) DHCP
b) Tunneling
c) NAT64
d) ICMP

Which field in the IPv6 header is used to identify packets belonging to the same flow?
a) Hop Limit
b) Flow Label
c) Source Address
d) Next Header
Which of the following is an advantage of SDN over traditional networking?
a) Static routing tables
b) Centralized control and programmability
c) Exclusive support for IPv6
d) Mandatory use of NAT for addressing
What type of actions can be defined in OpenFlow's "match + action" rules?
a) Only forward or drop packets
b) Modify, forward, drop, or send packets to the controller
c) Prioritize packets based on payload size
d) Split packets for fragmentation
In OpenFlow, flow table entries include all EXCEPT:
a) Matching fields
b) Associated actions
c) Packet counters
d) Transmission delays

Which middlebox improves website performance by storing frequently accessed content?
a) NAT device
b) Firewall
c) Cache
d) Load balancer
What is the role of a load balancer?
a) To drop packets during network congestion
b) To distribute traffic evenly across multiple servers
c) To manage IP address assignments
d) To prioritize video streaming traffic
What does "whitebox" hardware refer to in modern middleboxes?
a) Proprietary networking devices
b) Custom-built devices for a specific function
c) Generic hardware running programmable software
d) Equipment designed for IPv6-only networks
What happens when the MTU of a link is smaller than the IP datagram size?
a) The packet is dropped.
b) The datagram is fragmented.
c) The router requests retransmission.
d) The MTU size is increased dynamically.
What layer processes "datagram fragmentation"?
a) Transport layer
b) Network layer
c) Data link layer
d) Physical layer
What is the purpose of the "end-to-end argument" in Internet design?
a) To ensure that reliability is provided by the network core
b) To recommend implementing application-level intelligence
c) To mandate use of middleboxes for routing
d) To restrict packet switching to the physical layer
Which RFC defines the concept of the "end-to-end argument"?
a) RFC 3234
b) RFC 2475
c) RFC 1958
d) RFC 2131
How does buffer management in routers signal congestion?
a) By dropping low-priority packets
b) By marking packets with ECN or RED
c) By forwarding packets with reduced TTL
d) By increasing queuing delay for all packets
Answer: b) By marking packets with ECN or RED

The data plane is responsible for local forwarding at the router level.
True

Best-effort service in the Internet guarantees the order of delivery for packets.
False

The control plane uses distributed logic in software-defined networking (SDN).
True

Functions of the Network Layer

Forwarding and routing are identical processes in the network layer.
False

Routing is a global network function that determines packet paths.
True

The forwarding table in a router is used to classify and queue packets.
False

Packet scheduling ensures fair allocation of bandwidth among flows.
True

IPv4 addresses are represented in hexadecimal notation.
False

NAT allows multiple devices to share one public IP address.
True

The main reason for IPv6 adoption was the exhaustion of the IPv4 address space.
True

DHCP can return a network mask along with an assigned IP address.
True

The host part of an IP address is always determined manually.
False

A subnet mask helps identify the network and host portions of an IP address.
True

NAT increases the number of available public IP addresses.
False

Private IP addresses are routable over the Internet.
False

NAT allows devices in a local network to share a single public IP address.
True

IPv6 datagrams are larger than IPv4 datagrams by default.
True

IPv6 uses fixed-length headers to simplify processing.
True

IPv6 includes built-in support for QoS through its flow label field.
True

DN centralizes control logic using a remote controller.
True

OpenFlow is a protocol used for configuring flow tables in SDN switches.
True

Generalized forwarding is limited to network-layer headers.
False

"Match + Action" rules can include modifying packet headers.
True

Guaranteed delivery with fixed delay is an Internet service model.
False

The Internet's best-effort model ensures packets are delivered in order.
False

Round-robin scheduling always prioritizes the shortest queue.
False

Weighted fair queuing ensures bandwidth is divided proportionally among flows.
True

First Come, First Serve scheduling guarantees the shortest delay for all packets.
False

Firewalls are an example of middleboxes.
True

Middleboxes perform only routing functions.
False

NAT is a middlebox that changes packet headers to translate addresses.
True

IP fragmentation is only performed at the source host.
False

A router must reassemble all fragments before forwarding the datagram.
False

IPv6 was designed to replace IPv4 due to address exhaustion.
True

Flow tables in SDN switches are updated dynamically by controllers.
True

The time-to-live (TTL) field prevents indefinite packet circulation in the network.
True

Encapsulation in tunneling allows IPv6 packets to traverse IPv4 networks.
True

NAT traversal is a method to bypass address translation limitations.
True

OpenFlow uses "match + action" rules to configure network behavior.
True

IPv6 guarantees packet delivery over unreliable networks.
False